

The missing row: unrestricted internal diffeomorphisms, the protected phase of Lorentz-violating massive gravity, and a superselection theorem for frame dragging

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*Draft v0.8 — 2026-09-26 (v0.1 through v0.4 on 2026-07-21; v0.2 applied the 31 findings of the first in-house adversarial audit of the text, v0.3 the 12 findings of the second — record and both reports in `campana-m2/AUDITORIA-TEXTO-PAPER-2026-07-21.md`; v0.4 added the FRW section §9 from the verified task-4 campaign; v0.5 (2026-07-25) recorded L. Pilo's confirmation of the BCP misprint — email, 22 Jul 2026, cc D. Comelli, see §2 and `campana-m2/RESPUESTA-PILO-2026-07-22.md`; v0.6 applied the 2026-07-27 block-list-closure campaign — third text audit (30 findings), §9's own adversarial physics audit (one formula refuted and corrected in place: the attractor dispersion; the correction strengthens the freeze-out), the full quadratic FRW action (tensor confirmed, vector closed as a new identity, tensor-mass global factor closed as a convention), the $\{X,Y\}$ completeness proof (§3), the C5d reconciliation, the reference-verification pass, and the [16]–[18] content checks — record in `campana-m2/CIERRE-BLOCKLIST-2026-07-27.md`; **v0.7 integrates the state-selection campaign of 2026-07-27/28** — a superselection theorem decides the two-branch question that v0.1–v0.6 carried as the paper's central open caveat, in the direction that *suppresses* the paper's own falsifiable signature: §8 rewritten; abstract, scoreboard, §§1, 3, 6, 7, 9, 10, 11, Reproducibility, Novelty search and references updated; and the title changed accordingly (it read "...and a screened frame-dragging signature" through v0.6) — campaign record: three independent columns plus a final adversarial verifier with its own pipeline, `campana-m2/seleccion-estado/` — `SINTESIS-SELECCION-2026-07-28.md`, `verificador/VERIFICADOR-FINAL-2026-07-28.md`). **v0.8 (2026-09-26)**: the BCP exchange is now cited with permission — L. Pilo's email of 15 Sep 2026 grants it, confirms the k^2 sign of their (8.17) and comments on the sign convention of the tensor mass; formal reference [34] added and §2, §9, §10, scoreboard, acknowledgments and block list updated; the factor-2 point in their (8.18) stays on record as ours (three-route verified), with a clarification sent — record in `campana-m2/RESPUESTA-PILO-2-2026-09-15.md`. First written form of the results of the $m_2 = 0$ campaign (2026-07-20/21; campaign records and full verification chain in `especulativo/campana-m2/`: `HANDOFF-campana-m2-2026-07-20.md`, `VERIFICACION-BCP-2026-07-21.md`, `SECTOR-ESCALAR-2026-07-21.md`, `ACOPLE-MATERIA-PPN-2026-07-21.md`, `frw/FRW-2026-07-21.md`, `CIERRE-BLOCKLIST-2026-07-27.md`, `seleccion-estado/SINTESIS-SELECCION-2026-07-28.md`). Internal working paper of the speculative stage of the Intangible Field Theory (TCI) program, written to stand alone: nothing below depends on any other claim of that program. **Not submitted anywhere**. The five technical legs (dictionary, scalar sector, matter coupling, FRW, state selection) each passed an independent adversarial verification pass (independent re-derivation pipelines, including a blind refuter for the dictionary leg and a five-flank attack for the state-selection leg; no leg was refuted as a block — the FRW pass did refute and correct one published formula, declared in §9); this text has passed four audit passes (the fourth, over the v0.7 rewrite, on 2026-07-28); **what remains before any submission is on the block list at the end** (one final read of the assembled text, a ~250-word submission abstract, and the author's explicit sign-off). Honesty rules of the house apply: every assumption declared, every "exactly" backed by a reproducible computation, negative results stated with the same prominence as positive ones.*

Honest scoreboard

Each row states whose work each piece is.

Piece	Status	Whose
The phase $m_2^2 = m_3^2 = m_4^2 = 0$ of Lorentz-violating massive gravity exists, is UV-insensitive, and is healthy if $m_0^2 m_1^2 > 0$	Known since 2004	Dubovsky [1], eqs. (71)–(75)
The medium $\rightarrow$ graviton-mass dictionary (unitary gauge, self-gravitating media)	Known since 2016	Ballesteros–Comelli–Pilo (BCP) [4]
The operator $X + Y^2$ and the isentropic class $U(X + Y^2)$	Known since 2017 (thermodynamic route, no connection to the protected phase)	Celoria–Comelli–Pilo (CCP) [5]
Scalar-sector phenomenology of $\omega^2 \propto p^4$ media (Jeans time, r_c , t_c , bounds)	Known since 2003–2005	ghost condensate, ACLM [8, 9]
The missing row of BCP's Table 2: $\Phi^a \rightarrow \Psi^a(\Phi^b)$ unrestricted $\Rightarrow U(X,Y)$ — with a completeness proof: $\{X, Y\}$ exhausts the leading-order invariants (§3, machine-verified)	New (to the searches declared below; no prior classification of the unrestricted case found)	this paper
$U(X,Y)$ lands exactly on the protected phase: $m_2 = m_3 = m_4 = 0$ emerge from the vacuum conditions, no tuning, explicit ghost-free corner	New, machine-verified	this paper
Exclusion lemma: on-shell $m_1^2 = 2K_T$, $m_2^2 = -2G_T$ — protecting the graviton mass $\Leftrightarrow$ switching off the medium's shear phonons (at leading order)	New in this framework (the mass $\leftrightarrow$ rigidity link is known in other settings [17, 18])	this paper
The exact scalar dispersion $\omega^2(p)$ of this phase with gravitational mixing (Dubovsky's leading-order freeze-out made quantitative; the p^4 coefficient is NLO and UV-sensitive)	New (first explicit form)	this paper
Causal theorem: with conserved sources the leading-order solution is exactly GR; $\gamma_{PPN} = 1$ exact; the mass only selects the gauge	New for this phase (mechanism anticipated qualitatively by Dubovsky's "gauge-fixing" remark [1])	this paper
Frame dragging has two branches: relaxed medium $\Rightarrow$ Yukawa-screened (gravitomagnetic Meissner); co-rotating medium $\Rightarrow$ GR (up to $\delta \lesssim 10^{-41}$, below). The would-be signature — screened Lense-Thirring, with LARES-2 [23] giving $\Lambda < 1.12$ MeV on the relaxed branch — is suppressed by the superselection theorem of the next row: the phase predicts GR gravitomagnetism, and those bounds stand as counterfactuals of the unpopulated branch	New; no prior computation of rotating sources in any $m_1^2 \neq 0$ phase found	this paper
Superselection theorem: the internal charge of the residual symmetry has no spatial current ($P_a^i = 0$ — an exact off-shell identity, verified four independent ways), so its density is frozen pointwise, with the inversion corollary $P_a^0(x) = 0 \Leftrightarrow g_{0a}(x) = 0$. The two branches are disconnected superselection sectors; cosmological initial conditions populate the co-rotating (zero-charge) one, universally; NLO opens no backdoor ($\delta = -\hat{c}(\Lambda/\bar{M}_{Pl})^2$, ≥ 30 orders below sensitivity everywhere in the declared windows; charge leak $\leq 3 \times 10^{-14}$ over the age of the universe on a deliberately broken bound). Hypotheses declared, §8/§10	New, machine-verified (three campaign columns + adversarial verifier with its own pipeline)	this paper

Piece	Status	Whose
<i>FRW: the tensor masslessness is an FLRW identity (and so is the vector freezing — both potentials $\propto$ the Friedmann-II combination, from the full quadratic action); the protected Minkowski vacuum is the cosmological attractor (in the mirror phase it is not [10]); vacuum-form stress on the attractor ($w = -1$ whenever $U \neq 0$); transient = exact dust; scalar stays frozen — the attractor spectrum is purely damped, zero phase velocity (the earlier $\omega = \pm(H/\sqrt{2})k + 3iH$ was refuted by this paper's own 2026-07-27 audit and corrected in §9)*</i>	New as a package (mechanism from [8, 10], current known to [5] as entropy; §9 audited 2026-07-27 — one formula corrected, the correction strengthens the freeze-out; rest survived + the quadratic-action pass confirmed tensor and closed vector)	this paper
Misprint in BCP eqs. (7.6)–(7.7) — confirmed by L. Pilo [34] (email, 22 Jul 2026, cc D. Comelli, after re-checking their original notebooks; cited with permission granted 15 Sep 2026); two further misprints in BCP §8, eqs. (8.17)–(8.18) — the k^2 sign of (8.17) confirmed by L. Pilo [34] (15 Sep 2026), the factor 2 in (8.18) ours (three routes, §9); two misprints in Dubovsky eqs. (73), (99)	Found and cross-checked five independent ways (BCP (7.6)–(7.7)), then confirmed by L. Pilo; (8.17) confirmed by L. Pilo; (8.18) three-route verified, clarification sent to the authors	this paper
The Einstein–Hilbert kinetic term	Assumed, not derived (as in [1, 4, 5]; Weinberg–Witten out of scope)	—

No death criterion of the campaign was triggered. The price of the result — the exclusion lemma — and the suppression of the campaign's own falsifiable signature by its own theorem (§8) are stated with the same prominence as the wins: this draft carried "screened frame-dragging signature" in its title for six versions, and the same verification machinery that established the signature then established the theorem that suppresses it.

Abstract

In 2004 Dubovsky classified the phases of Lorentz-violating massive gravity and found one where the helicity- ± 2 graviton is **exactly massless, protected by the residual symmetry** $x^i \rightarrow x^i + \xi^i(\vec{x})$, UV-insensitive, and ghost-free when $m_0^2 m_1^2 > 0$ — and closed with the remark that his analysis "does not exhaust all possible subgroups of the diffeomorphism group." To the searches declared below, the medium realizing that phase was never constructed. Ballesteros–Comelli–Pilo (BCP) later built the general dictionary from self-gravitating media (four Stückelberg scalars Φ^A) to the five graviton mass parameters, classifying media by their internal symmetry — solids, fluids, superfluids, supersolids — all protected by (subgroups of) *volume-preserving* internal diffeomorphisms. We point out that their Table 2 has a missing row: media invariant under **unrestricted** internal spatial diffeomorphisms, $\Phi^a \rightarrow \Psi^a(\Phi^b)$ with no condition on the determinant. Two invariants survive at leading order, $X = C^{00}$ and the counterflow $Y = u \cdot \partial \Phi^0$, and the resulting medium $U(X, Y)$ lands exactly on the protected phase: the flat-space tadpole conditions force $m_2^2 = m_3^2 = m_4^2 = 0$ identically, with $m_1^2 = 2U_X \neq 0$ and an explicit ghost-free corner. The identification comes with a structural lemma: on-shell, $m_1^2 = 2K_T$ and $m_2^2 = -2G_T$, where K_T and G_T are the inertia and the rigidity of the medium's transverse (shear) sector — **the graviton mass is the shear rigidity of the medium**, so protecting the massless graviton is equivalent to switching off shear phonons at leading order: the medium must be a fluid. We then compute what this phase does in the lab. The scalar sector is frozen at leading order (a double zero, matching the content of Dubovsky's eq. 99); the $\omega^2 \propto p^4$ dispersion turns on only through next-to-leading-order operators, with a UV-sensitive coefficient, and with gravitational mixing the infrared is a slow Jeans instability — in that sector the medium is exactly a ghost condensate (dictionary $M^4 = 2m_0^2$, $\kappa = -\bar{M}^2/2$, in ACLM's notation), because m_1^2 cancels there. The sector where m_1^2 does not cancel is gravitomagnetism: with conserved sources the static and orbital predictions are exactly GR ($\gamma_{\text{PPN}} = 1$ exact, by a causal theorem: the mass term only selects the synchronous gauge and costs zero action on-shell), and

the vector sector has **two branches**. If the medium sits still relative to the distant stars, the gravitomagnetic potential obeys $(\nabla^2 - \mu^2)S_i = 16\pi\tilde{G}\tau_i$ — a gravitomagnetic Meissner effect, Lense–Thirring precession screened by $(1 + \mu a)e^{(-\mu a)}$ with $\mu = 2\Lambda^2/\tilde{M}_{\text{Pl}}$; if the medium co-rotates (a persistent-current state), the prediction is GR. Which branch nature picks turned out to be **decidable — and decided — by a superselection theorem**: the internal charge of the residual symmetry $\Phi^a \rightarrow \Phi^a + \xi^a(\Phi^\rightarrow)$ has no spatial current ($P_a^i = 0$, an exact off-shell identity of the full nonlinear theory, verified four independent ways), so its density is frozen pointwise, with the inversion corollary $P_a^0(x) = 0 \Leftrightarrow g_{0a}(x) = 0$ ($U_Y \neq 0$): the two branches are disconnected superselection sectors. Cosmological initial conditions — the FRW attractor of this same paper — populate the zero-charge, co-rotating sector, universally ($S = 0$ solves the source switch-on exactly for arbitrary history, moving sources included); the screened branch is unreachable, because the current that would transport the charge into it does not exist. Next-to-leading-order operators open no backdoor: the surviving correction is a universal renormalization of the gravitomagnetic coupling, $|\delta| = \hat{c}(\Lambda/\tilde{M}_{\text{Pl}})^2 \approx 10^{-61}\text{--}10^{-41}$ across the conservative viability window ($\leq 10^{-33}$ even at its 100 GeV extreme) — ≥ 30 orders of magnitude below current sensitivity everywhere in the declared windows — and the charge-leak rate is $\Gamma \cdot t_{\text{age}} \leq 3 \times 10^{-14}$, more than thirteen orders below one charge over the age of the universe, on an estimate deliberately broken against us. **The phase therefore predicts GR gravitomagnetism outright — not because the medium hides well, but because a conservation law forbids it to show** (under the hypotheses declared in §8: universal coupling, smooth Φ^a , the structural NLO basis). The screened Lense–Thirring signature that earlier versions of this work announced as the falsifiable prediction is thereby suppressed, and we say so with the same prominence: the LAGEOS/LARES bounds it would have implied ($\Lambda \lesssim 2.0\text{--}2.6$ MeV, and $\Lambda < 1.12$ MeV from the July-2026 LARES-2 measurement at its claimed 0.2% [verified here against its openly served Extended Data, Supplementary and peer-review material; main text pending]) now stand as counterfactuals of the unpopulated branch — observing a nodal deficit with the screened profile would falsify the theorem's hypotheses, not measure Λ — and the viability window for Λ is set by the scalar sector alone. To our searches (declared below), no computation of rotating sources — and no discussion of state selection of the Stückelberg medium around them — exists in any $m_1^2 \neq 0$ phase. On FRW backgrounds the story closes on its own: the tensor masslessness is an FLRW *identity* (X and Y are exactly blind to transverse-traceless strain); the density of the medium's conserved charge dilutes as a^{-3} and drives the background onto the protected corner (in the Minkowski basin) — the phase's Minkowski vacuum is a cosmological attractor, where its mirror-phase analogue is explicitly not; the attractor's stress is vacuum-form ($w = -1$ whenever $U^* \neq 0$) with a purely damped scalar spectrum, and the off-attractor transient redshifts as exact dust, reproducing the ghost condensate's cosmological-constant-plus-dark-matter phenomenology. As by-products, we report a misprint in the mass-parameter weights of BCP eqs. (7.6)–(7.7) — with the correct weights established five independent ways and since confirmed by L. Pilo [34] — two further misprints in BCP's §8, eqs. (8.17)–(8.18), the first also confirmed by L. Pilo [34] — and two minor misprints in Dubovsky's eqs. (73) and (99). All results are symbolic (SymPy) and public; individual scripts run in seconds to a few minutes.

1. Introduction

Massive gravity with broken Lorentz invariance is less constrained than its Lorentz-invariant cousin: with separate mass parameters $m_0\dots m_4$ for the different components of $h_{\mu\nu}$ (in the notation of [1–3]), the Boulware–Deser ghost can be evaded ($m_0 = 0$), the vDVZ discontinuity is absent, and — the case of interest here — there exists a phase where the spin-2 graviton is **exactly massless while the medium that breaks Lorentz invariance is still there**. (The ghost evasion via $m_0 = 0$ belongs to a different corner of the class; the

phase below is ghost-free through $m_0^2 m_1^2 > 0$ instead.) Dubovsky exhibited it in 2004 [1]: if the vacuum preserves the residual symmetry

$$x^i \rightarrow x^i + \xi^i(x^\gamma) \quad (\text{Dubovsky eq. 71})$$

then $m_2^2 = m_3^2 = m_4^2 = 0$ (his eq. 72) — and, unlike generic mass patterns, this one is stable under radiative corrections, because any counterterm must respect the residual symmetry (UV-insensitivity). The phase propagates the two GR tensor polarizations at the full speed of light, plus a vector sector controlled by m_1^2 and a scalar sector we discuss at length below. Its health condition is $m_0^2 m_1^2 > 0$ (his eq. 75). The empirical stakes of *exact* protection are set by the observational bound on the graviton mass, $m_g \leq 1.27 \times 10^{-23}$ eV [20]: a symmetry-protected zero satisfies it structurally, where any tuned small mass merely survives it. Twenty-two years of literature cite the phase; to our knowledge, and to the searches declared at the end, **nobody ever constructed the physical medium that realizes it**. Dubovsky's own closing line is an open invitation:

"Clearly, our analysis does not exhaust all possible subgroups of the diffeomorphism group. It is worth studying whether other interesting possibilities exist." [1]

The natural framework for the question is the self-gravitating-media program of Ballesteros, Comelli and Pilo (BCP) [4]: four derivatively-coupled scalars Φ^0, Φ^a (the medium's clock and comoving coordinates), a leading-order Lagrangian U built from the invariants of $C^A\{AB\} = g^{\mu\nu} \partial_\mu \Phi^A \partial_\nu \Phi^B$, and a dictionary from U to the five mass parameters in unitary gauge. Their Table 2 classifies media by internal symmetry: solids $U(\tau_n)$, perfect fluids $U(b, Y)$, superfluids $U(X, Y, b)$, supersolids (in the effective-field-theory-of-media tradition [14]) — every row protected by (a subgroup of) the internal diffeomorphisms that **preserve the volume element**, $\det(\partial\Psi/\partial\Phi) = 1$.

This paper adds the row that the table is missing — the medium invariant under internal spatial diffeomorphisms with **no restriction at all** — and shows that it is precisely the object Dubovsky's invitation asked for. We then push the phase through its first phenomenology with matter (to our searches): the result is a theory that imitates GR by theorem in the static and orbital regimes where GR is best measured — the radiative sector is open (§11) — and whose one computed candidate deviation, a two-branch gravitomagnetic response (itself new: all prior source computations in this class live in the mirror family $m_1^2 = 0$, and nothing exists for $m_1^2 \neq 0$ — searches declared below), is closed by a second, sharper result: a superselection theorem freezes the medium in the branch indistinguishable from Einstein (up to $\delta \lesssim 10^{-41}$, §8). The paper's observational content is therefore a clean structural prediction — GR in every static and orbital regime measured, and in gravitomagnetism, enforced by conservation laws rather than tuning (the radiative sector stays open, §11) — together with the two-branch structure and the theorem that decides it, both new.

Everything below is machine-verified symbolic algebra (SymPy; exact rational arithmetic; no truncations except the declared perturbative orders), reproducible in seconds to minutes from the public scripts (§ Reproducibility). The five technical blocks (dictionary + misprint; scalar sector; matter coupling; FRW cosmology §9; state selection §8) were each re-derived from scratch by an independent adversarial pipeline that shared no code with the original derivation — plus, for the dictionary leg, a blind refutation attempt, and, for the state-selection leg, a five-flank adversarial verifier with its own third pipeline; none refuted (the verifiers did catch minor slips — four in the scalar leg, one omitted term in the matter leg, and two bookkeeping blemishes in the state-selection leg (a script's closing print at odds with its own table, and one internal inconsistency about which bound governs) — all incorporated and on record).

2. Conventions, the calibrated dictionary, and a misprint in BCP

We work throughout in BCP's conventions [4]: signature $(-, +, +, +)$, unitary gauge $\Phi^0 = t$, $\Phi^a = x^a$, $M_{\text{Pl}} = 1$ unless restored, flat background $g = \eta + h$. The medium invariants are $X = C^{00}$ (background -1), $V^a = C^a\{0a\}$, the spatial block $B^{\{ab\}}$ (background $\delta^{\{ab\}}$), $Z^{\{ab\}} = V^a V^b$, $\tau_n = \text{Tr}(B^n)$, $y_n = \text{Tr}(B^n Z)$, $b = \sqrt{\det B}$, and $Y = u \cdot \partial \Phi^0$ (equal to $1/\sqrt{(-g_{00})}$ in the unitary gauge $\Phi^0 = t$) with u^μ the unit vector normal to constant- Φ^0 slices. The mass pattern is BCP's (7.3):

$$\sqrt{-g} U \supset t^{\{\mu\nu\}} h_{\{\mu\nu\}} + \frac{1}{4} [m_0^2 h_{00}^2 + 2m_1^2 h_{0i} h_{0i} - 2m_4^2 h_{00} h_{ii} + m_3^2 h_{ii}^2 - m_2^2 h_{ij} h_{ij}]$$

We built the complete dictionary $U(X, V, B, b, \tau_n, y_n, Y) \rightarrow (m_0^2, \dots, m_4^2)$ by direct second-order expansion (`masas_medio.py`) and calibrated it five independent ways: (i) all rotational-isotropy identities close; (ii) our m_0^2 reproduces BCP's (7.5) **exactly**; (iii) imposing the symmetry $\Phi^a \rightarrow \Phi^a + f^a(\Phi^0)$ reproduces Dubovsky's $m_1 = 0$ phase (his eq. 64); (iv) the gold calibration: the phonon Lagrangian derived by the Goldstone route (flat space, no unitary gauge — a computationally disjoint path) equals the mass Lagrangian under $h_{\mu\nu} \rightarrow \partial_\mu \pi_\nu + \partial_\nu \pi_\mu$ term by term, with **zero remainder** modulo total derivatives once tadpoles are imposed; (v) the transverse sector reproduces $L_T = \frac{1}{2} [m_1^2 (\partial_0 \pi^T)^2 - m_2^2 (\partial_i \pi^T)^2]$.

The misprint. Our dictionary disagrees with the *printed* BCP eqs. (7.6)–(7.7) in the weights of U_{τ_n} and U_{y_n} (flat background, $a = N = 1$):

	m_1^2 : U_{τ_n} weight	m_1^2 : U_{y_n} weight	m_2^2 : U_{τ_n} weight
this work	$-2n$ $(-2, -4, -6)$	$+2$ flat, including $n = 0$	$-2n(n+1)$ $(-4, -12, -24)$
BCP printed	-2 flat	$+2n$ $(0, +2, +4, +6)$	$-4(n+1)$ $(-8, -12, -16)$

Five independent lines of evidence select our weights: (1) the gold calibration above; (2) an exact-inverse-metric rederivation on directed configurations with the ε -series taken only at the end (`verificacion_independiente.py`); (3) a blind refutation attempt (an agent given only BCP's conventions and the dispute, no access to our files) that rederived the weights four ways and confirmed them — including the structural observation that δy_n is *exactly* uniform in n at $O(h^2)$ (Z is already $O(h^2)$), so the printed weight $2n \cdot U_{y_n}$ would annihilate the $n = 0$ term that BCP's printed sum explicitly includes; (4) internal consistency of BCP itself: their tensor-sector mass formula, their eq. (7.21) — arXiv v2 numbering throughout this paper, checked against the arXiv PDF on 2026-07-27 (the journal PDF was never accessible to us; their quadratic tensor Lagrangian is their (7.22)) — $M_2^2 = \Sigma n^2 U_{\tau_n}$ (quoted at $a = 1$; the full expression carries $a^{(-2(n-1))}$), is compatible with our weights and not with their printed (7.7); (5) the appendices of the later papers of the same group [5, 6, 7] use precisely our weights (flat in y_n from $n = 0$, n^2 in τ_n). No erratum of PRD 94, 124023 exists; the equations are unchanged between arXiv v1 and v2. **Confirmed by L. Pilo** [34]: the query letter was sent on 21 Jul 2026, and L. Pilo (email, 22 Jul 2026, cc D. Comelli), after re-checking the original Mathematica notebooks, confirmed the misprints in m_1^2 and m_2^2 and agreed with our expressions (cited with permission, granted on 15 Sep 2026; whether a PRD erratum will be filed is the authors' call — records in `campana-m2/RESPUESTA-PILO-2026-07-22.md` and `campana-m2/RESPUESTA-PILO-2-2026-09-15.md`). Caveats: the comparison was made at $a = N = 1$; the journal PDF itself was not accessible to us (paywall).

For the central result of this paper the issue is moot — $U(X, Y)$ depends on neither τ_n nor y_n — but the exclusion lemma of §5 does use those weights, so the verification chain above is part of this paper's evidence base.

Two minor misprints in Dubovsky [1] (found while cross-checking our scalar sector against his; neither affects his conclusions): in eq. (73), dimensional analysis requires Λ^2 where Λ^4 is printed (his strong-coupling scale — unrelated to the medium scale Λ introduced in §6; an “ $= 0$ ” is also omitted); in eq. (99), rederiving

from his own (86)–(88) gives $-6(\partial_0\tau)^2$ where $+6(\partial_0\tau)^2$ is printed (verified symbolically). Also for precision: the Einstein–Hilbert normalization of his section 5 is twice the canonical one, which rescales absolute mass conversions but no dimensionless condition.

3. The missing row: unrestricted internal diffeomorphisms and $U(X,Y)$

BCP's Table 2 asks of each medium: under which internal transformations of the comoving labels Φ^a is U invariant? Every row of the published table imposes volume preservation. Drop it:

$$\Phi^a \rightarrow \Psi^a(\Phi^b), \quad \det(\partial\Psi/\partial\Phi) \text{ unrestricted.}$$

A medium with this symmetry does not feel *any* deformation of its spatial labels: no shear, no compression, no dilation. At leading order in derivatives exactly two invariants survive:

- $X = C^{00}$ — the clock gradient squared (manifestly free of Φ^a);
- $Y = u \cdot \partial\Phi^0$ — with u^μ the unique direction invariantly defined by the flow lines ($u \cdot \partial\Phi^a = 0$, normalized). Y measures the **counterflow** between the clock condensate and the material flow lines.

Both invariances were verified *non-linearly* on a randomized exact instance each (one rational random metric, one quadratic internal diffeomorphism, exact arithmetic: $\Delta X = \Delta Y = 0$ exactly). That $\{X, Y\}$ **exhausts** the leading-order invariants is now proved and machine-verified (`campana-m2/exhaustividad_XY.py`, 19/19 checks in exact arithmetic), no longer inherited. The argument is elementary. At leading order (one derivative per field) the building blocks are the entries of $C^\wedge\{AB\}$ — $X, V^\wedge a, B^\wedge\{ab\}$ — plus the oriented Jacobian $Z_\varepsilon = \varepsilon^\wedge\{\mu\nu\rho\sigma\}\partial_\mu\Phi^0\partial_\nu\Phi^1\partial_\rho\Phi^2\partial_\sigma\Phi^3/\sqrt{-g}$, *the one scalar not a function of C (Dubovsky's general one-derivative action, eqs. (14)–(15) of [1])*. An internal diffeomorphism acts on these pointwise values only through its Jacobian $J = \partial\Psi/\partial\Phi/\{\Phi(x)\} \in GL(3)$ — already realized by linear Ψ ; derivatives of J enter only ∂C -operators, NLO by the derivative counting. On the physical domain (B positive definite, $\det C \neq 0$; the background sits inside) every (X, V, B) is brought to the normal form $(X, (\sqrt{W}, 0, 0), \delta)$, $W \equiv V \cdot B^{-1} \cdot V$, by $J = R \cdot A^{-1}$ with $B = AA^T$: the $GL(3)$ orbits are exactly the fibers of the pair (X, W) , so **every invariant is a function of (X, W)** — confirmed by the infinitesimal count (generic orbit dimension $8 = 9 - 1$, hence $10 - 8 = 2$ invariants). The Schur identity $\det C = \det B \cdot (X - W)$ gives $(C^{-1})_{00} = (X - W)^{-1} = -Y^{-2}$, i.e.

$$V \cdot B^{-1} \cdot V = X + Y^2,$$

with the sign of Y fixed by time orientation (u future-directed) as in [4]; the spatial block of the 4×4 inverse yields nothing new either ($V \cdot (C^{-1})|_s \cdot V = WX/(X - W)$). Z_ε itself is *not* invariant — $Z_\varepsilon \rightarrow \det J \cdot Z_\varepsilon$ with $\det J$ free — which is precisely how the volume-preserving column of the table dies (b, τ_n, y_n all transform), while $Z_\varepsilon/b = \pm Y$ adds nothing. Hence $U(X, Y)$, with no third argument. Two corollaries close the circle: restricting to $\det J = 1$, the same count gives three invariants $\{X, Y, b\}$ — BCP's superfluid row, recovered as a calibration — and dropping the clock leaves no invariant at all ($GL(3)$ acts transitively on positive B): a medium of Φ^a alone with the unrestricted symmetry is dynamically empty — the missing row *requires* the clock. The classification is the internal, time-independent analogue of the unitary-gauge operator classification of the EFT of inflation [27], whose larger residual group $\xi^i(t, \vec{x})$ leaves only g^{00} ; restricting to $\xi^i(\vec{x})$ is exactly what lets the counterflow Y survive. Scope declared: leading order only (the NLO basis of §6 is a separate statement); Wess–Zumino-type terms shifting by a total derivative are outside the algebraic argument (as in [1, 4]).

So $U(X,Y)$ is a two-component object — a clock condensate plus flow lines — whose only memory of ever having had a lattice is the flow direction: *a superfluid that forgot its lattice*. CCP [5] found the one-variable subclass $U(X + Y^2)$ in 2017 as the isentropic thermodynamic class — without noting the unrestricted

symmetry, and without the connection that is the point of this paper. What $U(X,Y)$ is **not**: it is not the ghost condensate $P(X)$ — flat-space tadpoles force $U_X = 0$ on $P(X)$ (hence its $m_1^2 = 0$), while $U(X,Y)$ sustains $U_X \neq 0$ because the counterflow compensates the tadpole ($U_Y = 2U_X$, next section); and it is not BCP's superfluid $U(X,Y,b)$ — any dependence on b regenerates $m_3^2 = U_{bb}/2 \neq 0$ and $m_4^2 \neq 0$. It is also not the finite-temperature superfluid $U(X,Y,b)$ of Khoury–Sakstein–Solomon [25], which realizes the *mirror* structure ($m_1^2 = 0$ identically for that class) and degravitates vacuum energy there; our row sits at $m_1^2 \neq 0$, which is where the two-branch structure and the superselection theorem of §8 live. $U(X,Y)$ is the exact point where the medium stops feeling its own volume.

4. The protected phase emerges from the vacuum conditions

Feeding $U(X,Y)$ to the calibrated dictionary (`parteB_medioXY.py`):

Off-shell:

$$\begin{aligned} m_0^2 &= -U_0/2 - 2U_X + 2U_{XX} - 2U_{XY} + U_Y/2 + U_{YY}/2 \\ m_1^2 &= U_0 + 2U_X \\ m_2^2 &= U_0 \\ m_3^2 &= U_0/2 \\ m_4^2 &= U_0/2 + U_X - U_Y/2 \end{aligned}$$

Flat-vacuum (tadpole) conditions: $t_{00} = -U_0/2 - U_X + U_Y/2 = 0$ and $t_{11} = U_0/2 = 0$, i.e. $U_0 = 0$ and $U_Y = 2U_X$. Then, **with nothing further imposed**:

$$\begin{aligned} m_2^2 &= 0, \quad m_3^2 = 0, \quad m_4^2 = 0 \quad (\text{identically}) \\ m_1^2 &= 2U_X \\ m_0^2 &= -U_X + 2U_{XX} - 2U_{XY} + U_{YY}/2 \end{aligned}$$

This is exactly Dubovsky's phase (72), and no accident: the residual symmetry (71), $x^i \rightarrow x^i + \xi^i(x^\rightarrow)$, is $\Phi^a \rightarrow \Psi^a(\Phi^b)$ read in unitary gauge. We verified directly that the resulting mass Lagrangian is invariant under $\delta h_{ij} = \partial_i \xi_j + \partial_j \xi_i$ with time-independent ξ : the masslessness is symmetry-protected, and any counterterm at leading order in derivatives that respects the symmetry is again of the form $U(X,Y)$, whose vacuum masses vanish again; higher-derivative counterterms (the NLO operators of §6) respect the symmetry too and do not regenerate the vacuum masses either — Dubovsky's UV-insensitivity, now with an explicit medium carrying it.

The ghost-free corner is explicit. Taking

$$U = (X+1) + 2(Y-1) + \frac{1}{2}(X+1)^2$$

($U(\text{vac}) = 0$, $U_X = 1$, $U_Y = 2U_X$, $U_{XX} = 1$), the dictionary gives

$$(m_0^2, m_1^2, m_2^2, m_3^2, m_4^2) = (1, 2, 0, 0, 0), \quad m_0^2 m_1^2 = 2 > 0 \quad \checkmark$$

satisfying Dubovsky's health condition (75). The vector sector has the healthy gapped dispersion of his eq. (73) whenever $m_1^2 = 2U_X \neq 0$ (that is the Goldstone-sector statement; for its fate with gravitational mixing see §6). The scalar sector is §6. Death criterion 1 of the campaign ($m_2^2 \neq 0$, or no ghost-free corner) was not triggered.

5. The exclusion lemma: the graviton mass is the medium's shear rigidity

The identification comes with a structural consequence, itself a result. From the transverse sector of the **general** medium (not just $U(X,Y)$), on-shell:

$$m_1^2 = 2 \cdot K_T, \quad m_2^2 = -2 \cdot G_T, \quad v_T^2 = m_2^2 / m_1^2$$

with $K_T = U_X + \Sigma_n U_{Y_n}$ the inertia of the medium's shear sector and $G_T = \Sigma_n n^2 \cdot U_{\tau_n}$ its **shear rigidity** (the gradient coefficient of the transverse phonon Lagrangian). Verified symbolically for the general U .

Lemma. At leading order, the medium's shear phonons propagate iff $m_1^2 m_2^2 \neq 0$. Hence *either* UV-insensitive phase of Dubovsky switches them off: the $m_1 = 0$ phase freezes them (no inertia); the $m_2 = m_3 = m_4 = 0$ phase leaves them rigidity-less ($v_T = 0$ — pure gauge of the residual symmetry (71)).

Moral. The spin-2 graviton mass *is* the shear rigidity of the medium, with sign and factor: $m_2^2 = -2G_T$. A medium with dynamical tensor order at leading order — a solid, a supersolid — endows the graviton with mass and loses the protection. Conversely, a medium that protects the massless graviton cannot contribute propagating spin-2 of its own at low energies: **protection and dynamical tensor order are mutually exclusive at leading order**. This is why solid inflation [16] generically has $m_2 \neq 0$ (tensor-mode masses from shear rigidity are a *feature* there), and it sharpens, from the gravity side, an intuition long present in other communities: in holographic massive gravity the bulk graviton mass maps to boundary elasticity [17], and in "crystal gravity" [18] solids and gravity are intertwined precisely through shear. What we add is the statement *inside* the BCP/Dubovsky framework, as an exact on-shell identity, with the iff, and with the consequence for protection.

Declared escapes (they delimit the claim rather than undermine it): higher-derivative kinetic structures (spin-nematic Goldstone routes à la Chojnacki [15], with non-relativistic dispersion), and induced gravity à la Sakharov — which is exactly the first non-claim declared in §10 (the EH term is assumed throughout, here as in [1, 4, 5]). Death criterion 2 did not trigger as a no-go: it became this lemma.

6. The scalar sector is frozen at leading order

Here this paper corrects its own campaign's earlier internal note, which had paraphrased Dubovsky as "one scalar mode with $\omega^2 \propto p^4$ ". The precise statement, now machine-verified with gravitational mixing included (independent adversarial re-derivation from $\sqrt{-g} R$; no refutation):

1. **At strict leading order ($U(X,Y)$ + quadratic EH), the scalar sector does not propagate.** The determinant of the scalar block is

$$\det \propto -(M_{Pl}^4 m_0^2 m_1^2 / 2) \cdot \omega^4 p^6$$

— a **double zero** at $\omega^2 = 0$ for every p . This matches the content of Dubovsky's eq. (99) (with the sign misprint noted in §2 corrected) and his verbatim conclusion ("mixing with gravity does not introduce new propagating modes").

2. **The $\omega^2 \propto p^4$ dispersion requires the medium's NLO operators** (the higher-derivative invariants compatible with the residual symmetry $\xi^i(x^\rightarrow)$: $\bar{K}_{ij}\bar{K}_{ij}$, $(\text{tr } \bar{K})^2$, ...). With them, exactly one scalar mode turns on, and its p^4 coefficient is **UV-sensitive**: $c_4 \approx -(\alpha + \beta)/m_0^2$, where α and β denote the couplings of the two quadratic invariants in $\bar{K}_{ij}$ — the perturbation of the extrinsic curvature of the constant- Φ^0 hypersurfaces — and σ, ρ those of the mixed NLO operators entering the scalar channel. (§7's static channel uses σ and ρ_{op} ; $\rho_{op} \equiv \rho$ — same operator, same normalization — was verified term by term on 2026-07-27, closing that question: `acople/estatico/C5d_dirimido.py`, checks [7a]/[7b]; see block-list item 7.) The

precise operator basis is fixed in the public scripts. The coefficient is *not* expressible in the masses alone. Dubovsky left the coefficient generic (verified against his source); the exact dispersion below is, to our knowledge, its first explicit form for this medium. The second root stays $\omega^2 = 0$ exactly at all orders computed — the surviving gauge zero after the double zero splits.

3. With gravity, the deep infrared is not p^4 but a Jeans p^2 : with $\kappa \equiv \alpha + \beta$ and $M_{\text{Pl}} = 1$,

$$\omega^2(p) = p^2 \cdot [\kappa(2p^2 - m_0^2) + p^2 m_0^2 (\rho^2/2 - 2\kappa\sigma)] / [m_0^2(-2 - \alpha + 3\beta + \alpha^2 + 3\alpha\beta)]$$

whose EFT limit ($|\alpha|, |\beta| \ll 1$) is $\omega^2 \approx (\kappa/2)p^2 - [\kappa/m_0^2 + (\rho^2 - 4\sigma\kappa)/4]p^4$ and whose decoupling limit $M_{\text{Pl}} \rightarrow \infty$ recovers the pure p^4 form. The no-ghost conditions, disaggregated (refining $m_0^2 m_1^2 > 0$): $\kappa < 0$ (ghost-condensate sign of the NLO), $\mathbf{m}_0^2 > 0$ (no gradient instability at p^4), $\mathbf{m}_1^2 > 0$ (healthy vector). In the explicit corner $(m_0^2, m_1^2) = (1, 2)$ with $\beta = -\varepsilon$ (a small positive parameter — unrelated to the perturbative bookkeeping ε of §2): kinetic coefficient $K = 3 + 2/\varepsilon > 0$ and $\omega^2 = \varepsilon p^2(2p^2 - 1)/(2 + 3\varepsilon)$ — a soft Jeans below $p^2 = 1/2$, stable above.

4. In this sector the medium is *indistinguishable* from the ghost condensate. The dictionary is exact and arithmetic (residue 0 by two routes): $M^4 = 2m_0^2$, $\kappa = -\tilde{M}^2/2$ in ACLM's notation [8]. The structural reason: $\mathbf{m}_1^2$ **cancels exactly in the scalar dispersion and in the scalar kinetic term** — it lives only in the vector sector. All of ACLM's phenomenology therefore imports wholesale: the Jeans instability is harmless ($\tau \approx 18\text{--}21$ Hubble ages at $\Lambda = 10$ MeV, $\propto \Lambda^{-3}$; Hubble friction erases it below $\Lambda \approx 30$ MeV; above, nonlinearities saturate it in percent-level lumps), and the viability window for $U = \Lambda^4 \hat{U}$ runs from $\Lambda \sim 10^{-3}$ eV (where the medium would *be* the dark energy) up to $\Lambda \lesssim 10$ MeV conservatively, ~ 100 GeV if percent-level lumping is accepted [9]. The hierarchy $\Lambda \ll M_{\text{Pl}}$ is declared, not explained — the standard ghost-condensate tuning, inherited, not cured.

The remaining sectors with gravity, verified: the vector does not propagate at LO — $\det = (m_1^2/4) \cdot \omega^2 p^2 (M_{\text{Pl}}^2 + \alpha)$; the gapped dispersion (73) quoted in §4 is the Goldstone-sector result, while with gravitational mixing the vector is a constraint at this order; and solving that constraint with sources is exactly what §8 does. The tensor is massless **exactly**, with $c_{\text{T}}^2 = M_{\text{Pl}}^2/(M_{\text{Pl}}^2 + \alpha)$: its masslessness is untouched by everything above, while $\alpha \neq 0$ shifts its speed — GW170817's $|c_{\text{T}} - 1| \lesssim 10^{-15}$ therefore bounds α/M_{Pl}^2 , a constraint we flag for the phenomenological pass rather than evaluate here. Mode counting: 2 (LO), 2+1 (NLO), matching Dubovsky.

The one thing the ghost condensate is *forbidden* to have — by its own residual symmetry, verified textually — is $m_1^2 \neq 0$. That is the one sector where this medium is structurally distinct from the ghost condensate — and §8 shows that what the distinction buys is a conservation law, not a signature. It is the subject of the next two sections.

7. Matter coupling: Einstein by theorem, not by tuning

Couple a conserved source $T_{\mu\nu}$ minimally to $g_{\mu\nu}$ (the medium couples only gravitationally; no direct Φ -matter coupling — declared assumption). One expects a massive theory to deform Newton. It does not, and the reason is structural:

Causal theorem (the causal candidate satisfies both the four massive-system equations and the four pure-GR equations — residues [0,0,0,0] in each set). For any conserved source switched on at a finite time, the leading-order solution of the massive system is **exactly** the solution of pure GR; the mass term selects the synchronous gauge of that solution and costs zero action on-shell ($L_{\text{mass}} = 0$ identically on the causal branch). Consequently $\gamma_{\text{PPN}} = 1$ **exactly** (not approximately) in all principal regimes, Newton's constant is the bare one (no renormalization), and the static potential of an eternal source — the oscillating

$\cos(\mu_s r)/r$ family, $\mu_s^2 = m_0^2/(2M_{\text{Pl}}^2)$, *not* Yukawa — is a fixed point that a causal history approaches only on times $t_c \sim 18$ Hubble ages at $\Lambda = 10$ MeV (scaling as Λ^{-3} , §6 — unreachable across the conservative window $\Lambda \lesssim 10$ MeV), exactly as in the ghost condensate [8], with which the dictionary is again arithmetic ($\mu_s^2 = m_{\text{ACLM}}^2$). At NLO the static PPN correction is

$$\gamma(p) = 1 - 2\sigma p^2/M_{\text{Pl}}^2 + p^4[\rho_{\text{op}}^2/(M_{\text{Pl}}^2 m_1^2) + 4\sigma^2/M_{\text{Pl}}^4] + O(p^6)$$

— scale-dependent, outside the constant- γ PPN book-keeping, and parametrically far below solar-system sensitivity for the allowed Λ . The closed-form causal initial-value problem confirms the theorem dynamically: $\Delta\Psi(t,p) = A(p)[1 - \cos \Omega(p)t]$, with $\cos \rightarrow \cosh$ in the Jeans band. (Dubovsky's remark that the mass here "has the form of a gauge fixing" [1] finds its precise content in the vector sector: §8.)

In every static and orbital regime computed here, then, this phase *is* Einstein — by theorem, not by calibration. (The radiative sector — binary-pulsar damping — is not computed in this paper; see Open problems.) A deviation, if any, must therefore be sought in the one sector this theorem does not cover — and there, §8 finds a second theorem waiting.

8. Frame dragging: two branches, and the superselection theorem that picks one

The exception to §7's theorem is the sector where m_1^2 lives: gravitomagnetism. Here the campaign found structure that, as far as our searches reach, nobody had noticed in any phase of this class — first a two-branch response, then the conservation law that decides between the branches. Through v0.6 this paper carried the decision as its central open caveat and the screened branch as its falsifiable signature; the 2026-07-27/28 state-selection campaign closed the question, in the direction that suppresses the signature. We keep the full two-branch structure on record — it is the stage on which the theorem acts, and the counterfactual it kills is what makes the prediction sharp.

The two branches. The same residual symmetry $\xi^i(\vec{x})$ that protects $m_2 = 0$ gives the vector sector an exact conservation law, which leaves a flat direction in the space of stationary vector states. The physical consequences bifurcate:

- **Relaxed branch (medium at rest relative to the distant stars — the naive ground state).** The gravitomagnetic potential $S_i \equiv h_{0i}$ obeys a **derived** massive equation sourced by the transverse momentum density τ_i (the T_{0i} of the rotating source),

$$(\nabla^2 - \mu^2) S_i = 16\pi\tilde{G} \tau_i, \quad \mu^2 = m_1^2/(M_{\text{Pl}}^2 + \alpha), \quad \tilde{G} = G \cdot M_{\text{Pl}}^2/(M_{\text{Pl}}^2 + \alpha)$$

— a gravitomagnetic Meissner effect: the medium expels frame dragging beyond the screening length $\ell_1 = 1/\mu$. For a rigidly rotating sphere of radius R (exterior, exact):

$$h_{0i} = 2\tilde{G} (\vec{x} \times \vec{J})_i (1 + \mu r) e^{(-\mu r)} / r^3 \cdot \mathfrak{F}(\mu R), \\ \mathfrak{F}(y) = 15[(y^2 + 3)\sinh y - 3y \cosh y]/y^5,$$

with the $\mu \rightarrow 0$ limit reproducing linearized Kerr exactly. Nodal precession of a satellite at semi-major axis a is suppressed by $\mathcal{S} = (1 + x)e^{(-x)}$, $x = \mu a$ — note **no linear term** (the Yukawa dipole cancels it); the gyroscope factor — GP-B's observable [28] — is $(1 + x - x^2)e^{(-x)}$. $\vec{\Omega}_{\text{node}} \parallel \vec{J}$ holds for arbitrary inclination.

- **Co-rotating branch.** The exact solution at leading order is **GR in disguise**: the medium co-rotates with the frames and the screening is invisible (the NLO shift it does acquire is quantified below). Its Hamiltonian lies above the relaxed branch's (verified, robust to boundary terms) — but the comparison, as the theorem below shows, is between states of *different superselection sectors*: within its own sector

the co-rotating state is the unique stationary state, i.e. that sector's ground state. Dubovsky's remark that the mass term here acts as a gauge fixing [1] describes this branch; the relaxed branch, and the choice between them, are new.

A bookkeeping note on the co-rotating solution (from the adversarial pass): in unitary gauge it shows a linearly growing $h_{xz} \propto t$. This is pure label winding — a shear deformation that $U(X,Y)$ does not see (the gauge-invariant transverse profile $\sigma \equiv S - \dot{F}$ stays constant) — removable by passing to physical coordinates, where the state is exactly linearized Kerr for all t ; the large- t description belongs in those coordinates.

The superselection theorem. The residual symmetry protects more than the graviton mass. For $\sqrt{-g} U(X,Y) + EH$, the Noether current of the internal transformations $\Phi^a \rightarrow \Phi^a + \xi^a(\Phi^{\rightarrow})$ has **no spatial components**:

$$P_a^i = 0 \text{ identically, } P_a^\mu \equiv \partial(\sqrt{-g} U)/\partial(\partial_\mu \Phi^a)$$

— an *off-shell algebraic identity* of the full nonlinear theory, exact at every order in $\hbar$. The reason is structural: the finite $GL(3)$ action on the Gram matrix of the $\partial\Phi^A$ leaves Y invariant as a statement of block linear algebra, so the infinitesimal corollary $P_a^\mu \partial_\mu \Phi^b = 0$ holds off-shell, and in unitary gauge ($\partial_\mu \Phi^b = \delta^b_\mu$) it collapses to $P_a^\mu = 0$ with no computation left. Verified **four independent ways**: an exact solve- u elimination (charge column), a Levi-Civita closed form (vertex column), a third parametrization by Gram determinants (the adversarial verifier's own), and — at the cubic order where the first backdoor would live — an explicit vertex-by-vertex expansion (vertex column: all 36 components of the $\hbar^2 \partial_i \pi$ vertices vanish; the spatial current with the Goldstone excited is pure advection, slaved to the density — charge rides *with* the medium, never *through* it). The transformation leaves $g_{\mu\nu}$ and matter untouched, so the current has no Einstein–Hilbert or boundary piece ($\delta L_{EH} = \delta L_{mat} = 0$ identically, $K^\mu = 0$): nothing is omitted. On-shell, $\partial_t P_a^0 = 0$ **pointwise** — the charge density is frozen where it sits, with closed form

$$P_a^0 = \sqrt{-g} \cdot U_Y \cdot g_{0a} / (g_{00} \sqrt{-g_{00}}) = -m_1^2 h_{0a} + O(\hbar^2)$$

and, since the prefactor never vanishes ($U_Y \neq 0$, regular metric), with an **inversion corollary**: $P_a^0(x) = 0 \Leftrightarrow g_{0a}(x) = 0$, pointwise and exact in $\hbar$. The sector label is the full shift profile in comoving coordinates — the separation is non-perturbative and field-complete, not an artifact of a single mode or of linear order. The two branches carry different charge distributions: on the verified mode, $C = -m_1^2 S$ (an Einstein-constraint identity), with $C = 0$ exactly on the co-rotating branch (in *its own* coordinates the medium sees no shift) and $C \neq 0$ mode by mode on the relaxed one (a dipolar distribution, zero net charge). Passing from one to the other would transport charge between points — **and the current that would do it does not exist**. The branches are disconnected superselection sectors (we use "superselection" in the dynamical sense: no operator of the theory transports the charge between points, so no physical process connects the sectors). The escape "relaxed profile plus a compensating charge cloud" fails too: the constrained system {constraint, $C = 0$, $\sigma = \sigma_{relaxed}$ } — σ the gauge-invariant transverse profile defined above — has no solution (verified) — the co-rotating state is the *only* stationary state of the zero-charge sector. Its "extra energy" cannot be released either: on the verified mode $H(C; \tau) = C^2/(4m_1^2) + (\tau - C)^2/(4Ap^2)$ — A the modal kinetic coefficient ($\approx \tilde{M}_{PI}^2$ in the campaign's conventions), p the mode wavenumber — is minimized by the relaxed charge, but lowering H requires changing C — no channel.

Population is universal. Which sector does nature fill? $S = 0$ solves the source switch-on problem **exactly** for arbitrary source history $\tau(t)$ — fast, slow, non-monotonic; the plane-mode dynamics is first order in time (zero eigenfrequencies), so the state is labeled by the charge alone (closed-form general solution; RK4 control across switch-on times spanning six decades: $|S_{final}/S_{relaxed}| \sim 10^{-13}$, consistent with integrator error — the symbolic solution is exactly zero) — and even for a source that **moves while it grows**: with $T^{0x} = \tau(t)\cos(p(z - z_0(t)))$, $\tau(t)$ and $z_0(t)$ arbitrary and the stress completed by conservation, $S \equiv 0$ solves all four

equations exactly (two quadratures; the matter transports its own momentum, the medium stays at zero charge). Accretion does not populate the relaxed branch. Cosmological initial conditions supply exactly this zero: on FRW, $P_a^0 \propto g_{0a} = 0$ identically at every epoch — §9's vector-freezing identity is this same structure — and any primordial charge dilutes as a^{-3} . Earth's non-axisymmetric, time-dependent rotation cannot pump the mode either: the response has no poles (transfer function bounded, no resonance at $2\Omega \oplus$ and no non-resonant secularity; symbolic $S = 0$ exact, RK4 over 200 revolutions: $\max|S| \sim 3 \times 10^{-11}$, integrator-level).

The formation of a planet leaves the medium co-rotating. Always. The state that through v0.6 we called "excited" is the one the universe prepares — and the theorem forbids it to decay.

NLO opens no backdoor. Higher-derivative operators could in principle add a spatial piece to the current ($\partial L / \partial(\partial \Phi) \cdot \partial \xi$). The operators available to them are the invariants of the flow u (expansion, shear, vorticity, acceleration: $\theta, \sigma_{ij}, \omega_{ij}, a_i$) — a structural statement (every $GL(3)$ -invariant touches $\partial \Phi^a$ only through u , and u depends on the Goldstone only through $\dot{\pi}$, so every internal spatial current carries a time derivative up front), declared as such rather than a formal completeness theorem (§10, §11). Within that basis: the new current term is a total time derivative — zero exactly on stationary states, and zero *modal-linear leak* exactly even in transients; the corrected charge is conserved for arbitrary $\tau(t)$; a two-mode NLO run shows no inter-mode leak at quadratic order (the modal Lagrangian separates; inter-mode coupling starts at cubic order, where the exact all-orders theorem already covers the charge). The (nonlinear) $\times$ (NLO) leak estimate — a dimensional chain, declared as such, built from the actual drives ($v_{\text{drag}} \approx 7.2 \times 10^{-16}$ at Earth's surface; the $J_{22} \approx 1.8 \times 10^{-6}$ forcing at $2\Omega \oplus$; pole-free response; cubic rectification) — gives $\Gamma \cdot t_{\text{age}} \leq 3.5 \times 10^{-35}$, and $\leq 2.7 \times 10^{-14}$ on a version of the bound *deliberately broken against us* (oscillating flux counted as coherent drain): more than thirteen orders of magnitude below one charge over the age of the universe even on the broken bound (thirty-four on the structural one). **The co-rotating sector does not drain.** What NLO does do is shift the co-rotating state off exact GR: the surviving correction is a **universal renormalization of the gravitomagnetic coupling** — momentum-independent at leading order, no Yukawa resurrection —

$\delta_{\text{ret}} = -\hat{c} (\Lambda / \tilde{M}_{\text{Pl}})^2$; $|\delta_{\text{ret}}| = 1.7 \times 10^{-61} \dots 1.7 \times 10^{-41}$ at $|\hat{c}| = 1$ (either sign) across the conservative window $\Lambda = 10^{-3} \text{ eV} \dots 10 \text{ MeV}$ — the campaign's numerical scan ran to $\Lambda = 1 \text{ MeV}$, where $|\delta_{\text{ret}}| \leq 1.7 \times 10^{-43}$, and the formula extends monotonically; even at §6's 100 GeV twinkling extreme it reaches only 1.7×10^{-33} . ($\hat{c}$ is the $O(1)$ UV-sensitive NLO coefficient; conventions as in the Numbers paragraph.)

LARES-2's 2×10^{-3} sensitivity sits ≥ 38 orders away across the conservative window, ≥ 30 even at the 100 GeV extreme (reviving observability would need a UV cutoff near 10^{17} GeV — $\hat{c} \sim 10^{40}$ at $\Lambda = 1 \text{ MeV}$, $\sim 10^{30}$ at 100 GeV, the same cutoff either way — inside a medium whose own cutoff is at the Λ scale; the NLO shift of the relaxed branch is $< 10^{-21}$ across the campaign's full scan, so the branches stay sharply distinct). This δ_{ret} settles what would otherwise be the open problem of NLO frame dragging on the co-rotating branch: the first observable distinguishing the persistent current from GR proper *exists*, and is unobservable by thirty-plus orders of magnitude. No pathology accompanies it: the modal Hamiltonian stays positive across the EFT window for either sign of $\hat{c}$, the flat direction survives as $\omega^2 = 0$ (not $\omega^2 < 0$), and the one apparent Ostrogradsky ghost (from the acceleration operator) is an artifact of truncation, removed by standard order reduction — and in any case none of the stationary branches depends on that operator (verified with generic coefficient, no order reduction).

The remaining channels die by imported physics. (i) *Vortices / phase slips of Φ^a* — the one genuinely non-perturbative escape, since a non-smooth Φ^a evades the theorem — face the standard persistent-current barrier physics [32]: the half-ring barrier — the maximum of $E_R - P_R \cdot v$ over ring size — scales as $1/v$, **divergent at low drive**, and the drive here is $v_{\text{drag}} \sim 10^{-15}$; persistent currents in superconductors are

measured stable beyond $\sim 10^5$ yr, and superfluid rings far below criticality decay only near v_c (the framework's declared limitations near v_c are irrelevant at $v/v_c \sim 0$ [32]). There is also no known thermal bath to activate them. (ii) *Scalar Cherenkov* — the one gapless channel — carries energy but no charge (it cannot change the sector label), and its rate is imported and negligible: the moving-source problem of the ghost condensate was solved in 2004–2007 — the wake structure [29], the co-moving observer recovering Newton [30], and the energy-loss rate $\dot{E} \sim \alpha M^2 v^3$ of [9] (their α and M — the condensate scale — not the NLO couplings of §6; textually "completely negligible for macroscopic objects", with their own Sun estimate ~ 20 W, reproduced here as 24.3 W). Applied to Earth's rotating non-axisymmetry (a declared proxy: the translational rate evaluated on the rotational drive), the implied spin-down is $\leq 2.8 \times 10^{-15}$ W at $\Lambda = 1$ MeV against a free margin of $\sim 3.6 \times 10^{11}$ W — the campaign's declared allowance of 10% of Earth's measured tidal-dissipation budget (provenance in `a4_numeros.py`) — 26 orders. (iii) *Superradiance of the gapped sector* is kinematically forbidden: with gravitational mixing the vector sector propagates no mode at all at this order (§6), and in the Goldstone-sector reading its modes are gapped at m_1 — either way, nothing exists at $\omega = m_{az} \Omega \ll m_1$ to absorb into (m_{az} the azimuthal number). (iv) *UV tunneling* — "hole" nucleation in the condensate [33] — has a rate that is exponentially sensitive to the UV completion: inherited with the ghost-condensate scalar sector, declared as an uncertainty, not computed in the EFT. One structural point closes the circle: the ghost condensate itself could never face our branch question, because its flow is **irrotational** ($u \propto \partial\varphi$ [9]) — it *cannot* co-rotate. Co-rotation requires the vector sector that $m_1^2 \neq 0$ provides; the question, and the theorem that answers it, exist only in this row. That is also why twenty-two years of ghost-condensate phenomenology never met it.

Numbers, and what the measurement now means. Restoring units, $\mu = 2\Lambda^2/\bar{M}_{Pl}$ with $\bar{M}_{Pl}$ the reduced Planck mass — the conventions of the coupling-campaign scripts: Λ is normalized by $U = \Lambda^4 \hat{U}$ with $\hat{U}_X = 1$ at the vacuum (so $m_1^2 = 2\Lambda^4$), and $M_{Pl}^2 \equiv 1/16\pi G = \bar{M}_{Pl}^2/2$. (The state-selection scripts normalize $m_1^2 = 4\Lambda^4$ instead; the $O(1)$ rescaling is absorbed in the UV-sensitive $\hat{c}$ and changes no number quoted here — checked.) I.e.

$$\ell_1 = 2.40 \times 10^8 \text{ m} \cdot (\text{MeV}/\Lambda)^2 \quad \text{— at } \Lambda = 1 \text{ MeV, roughly the Earth–Moon distance.}$$

Had the relaxed branch been the populated one, Lense–Thirring measurements with LAGEOS/LARES (accuracies claimed between $\sim 5\%$ [21] and $\sim 2\%$ in the same team's 2019 update [21b]; the error budget is contested by independent assessments [22]) would give $\Lambda \lesssim 2.0\text{--}2.6$ MeV, and the **July-2026 LARES-2 measurement** [23] — LARES-2 (in orbit since 2022) combined with LAGEOS and GRACE, agreement with GR reported at the 0.2% design accuracy — would push that to $\Lambda < 1.12$ MeV, i.e. $\ell_1 > 30 R_\oplus$, with a predicted nodal deviation of 0.13% at $\Lambda = 1$ MeV (semi-major axis $a = 1.227 \times 10^7$ m, as in the campaign scripts). **By the theorem above, those numbers are counterfactuals of a branch nature does not populate.** For this phase the measurement is a passed consistency test — predicted deviation $|\delta_{ret}| \leq 1.7 \times 10^{-41}$ across the conservative window ($\leq 1.7 \times 10^{-43}$ over the campaign's scan) — not a bound on Λ : the viability window is set by the scalar sector alone (§6: $\Lambda \lesssim 10$ MeV conservatively, ~ 100 GeV with percent-level lumping [9]). The counterfactual retains one sharp use: a nodal deficit with the specific screened profile $\mathcal{S} = (1 + \mu a)e^{(-\mu a)}$ would now *falsify the theorem's hypotheses* (§10), not measure Λ . The error-budget dispute — [22] contests the 0.2% via the even-zonal cancellation's dependence on achieved injection accuracies; the team's public Peer Review File carries the quantitative reply (injection $\sim 15\times$ better than the pre-launch assumption, inclination sum 180.01° , even-zonal residuals in the Monte Carlo), alongside [24]; [23] itself does not cite [22] — therefore no longer converts into MeV-scale physics for this phase; it matters for the counterfactual's strength only. (An earlier internal bound used a criterion with a spurious linear term; the corrected suppression factor weakens the counterfactual bound by $2.7\times$ in Λ — the correction is part of this paper's record. [23] is verified here against its openly served primary material: abstract, reference list, Extended

Data Tables 1–3 — the 0.2% is the Total-RSS of Table 2(b), with the frame-dragging parameter $\mu_{LT} = 1.0001 \pm 0.0019$ (their notation — unrelated to our screening mass μ) and the LARES-2 semi-major axis 12 264 567 m — which rounds to 1.2265×10^7 , 0.04% off the scripts' input 1.227×10^7 m, negligible at the 0.13% level — the Supplementary Information, and the 88-page Peer Review File. The paywalled main text (pp. 332–335) remains unread and stays on the block list.)

Hypotheses of the theorem (all declared in this paper — the smoothness of Φ^a , implicit in any EFT of smooth fields, is made explicit here): (1) $L = \sqrt{-g} U(X,Y) + EH$ at leading order, with NLO built from the flow invariants — the basis $\{\theta, \sigma, \omega, a\}$ as a structural argument, not a formal completeness theorem (§11); (2) universal coupling — matter does not touch Φ^a directly (assumed throughout, §7); (3) Φ^a smooth — no vortices; if nucleated anyway, the barrier physics above applies; (4) $U_Y \neq 0$ and a regular metric (for the inversion corollary); (5) negligible primordial vector perturbations (§9's identity plus a^{-3} dilution; a conspiracy would need the dipolar profile *correlated with the source*). Verification chain: three independent campaign columns (A: vertices/NLO/numbers; B: charge/population; C: primary-source bibliography — the `colA/colB` scripts of § Reproducibility) plus a final adversarial verifier with its own third pipeline — different parametrization of Y , different EH construction — that reproduced every column run and attacked the theorem on five flanks without breaking it; scripts and records in `campana-m2/seleccion-estado/`.

Prior literature (searches declared). Every computation of sources we could find in Lorentz-violating massive gravity lives in the *mirror* family $m_1^2 = 0$ with massive tensor: the static solutions of [10, 11, 12], and the rotating "bumpy black holes" of [13], which state verbatim that the linear-in- J dragging is exactly GR there. PPN α_1, α_2 appear nowhere in the canonical review [3], and our searches found no such computation for any of Dubovsky's phases; our scale-dependent screening is in any case outside the constant-coefficient PPN formalism. Full-text searches for screened/Yukawa Lense–Thirring in massive or Lorentz-violating gravity return nothing comparable; the closest conceptual echo is the gravitomagnetic London moment once claimed for laboratory superconductors [19] — an unconfirmed, never-replicated experimental claim, and in any case a different object in a different setting. The Einstein-aether/khronometric family modifies frame dragging *fractionally* (via preferred-frame parameters), without a screening length: different in kind. A dedicated sweep for the state-selection question itself (metastability of Stückelberg-medium configurations around rotating sources in $m_1^2 \neq 0$ phases; seven queries, declared in the campaign record) returns nothing; the moving-source literature lives entirely in the irrotational scalar sector [29, 30, 9], which cannot co-rotate. One neighboring headline is cited for completeness: a "ghost condensate model of flat rotation curves" [31] — a static, monopolar, spatial-gradient mechanism, unrelated to rotation or to the vector sector. The two-branch structure, and the superselection resolution, are — to the searches declared below — new.

9. Cosmology: the protected corner is a cosmological attractor

(This section entered in v0.4, after the two audit passes of §§1–8; it received its own adversarial physics audit on 2026-07-27 — one leading-order formula was refuted and corrected (the attractor dispersion, below), the rest survived independent re-derivation; record in `campana-m2/CIERRE-BLOCKLIST-2026-07-27.md` §5 and the correction addendum of `campana-m2/frw/FRW-2026-07-21.md`. Full record: `campana-m2/frw/`.)

On an FRW background ($ds^2 = -dt^2 + a^2 dx^2$, unitary gauge $\Phi^0 = \varphi(t)$, $\Phi^a = x^a$) the medium's invariants collapse to $X = -\dot{\varphi}^2$, $Y = \dot{\varphi}$ — the background lives on the Schur surface $X + Y^2 = 0$. Three statements, each machine-verified by an independent adversarial pipeline (three derivation routes for the background; two routes and both polarizations for the tensor; frozen-coefficient perturbation analysis for the scalar):

The tensor protection is an FLRW identity. A transverse-traceless perturbation leaves X and Y invariant *exactly, to all orders in h* — in fact $\delta X = \delta Y = 0$ holds for *any* purely spatial perturbation in synchronous gauge, stronger than TT — so $U(X,Y)$ touches the tensor sector only through $\sqrt{-g}$: the tensor mass vanishes

identically on any spatially flat FLRW solution of the theory — no vacuum conditions required; with additional perfect-fluid matter the standard GR cancellation applies (verified here for the self-sourced case) — with $c_T^2 = 1$ at leading order (the NLO speed shift α of §6 is not recomputed on FRW) and the standard 3H friction ($H \equiv \dot{a}/a$). Re-derived independently from the full quadratic action (2026-07-27, `frw/accion_cuadratica_frw.py`): the tensor mass coefficient is proportional to $U + 2\ddot{a}/a + H^2$, which vanishes by Friedmann II alone — and the same pass settled the apparent global factor 2 against BCP's tensor-mass formula (their (7.21), arXiv v2 numbering): their printed formula is **correct**, the factor was their $M_{Pl}^2 = 1/16\pi G$ convention (no $1/2$ on R) versus ours (`frw/factor_722.py`, 17 checks). **The vector sector is frozen on any FLRW too:** from the same quadratic action, the vector potential is proportional to the very same Friedmann-II combination — identically zero on-shell for arbitrary $U(X,Y)$ and $a(t)$ — leaving a nonpropagating constrained mode (the exact FLRW generalization of the Minkowski double zero of §6), verified by three routes and, term by term on-shell, against BCP's printed vector-sector Lagrangian (their (7.23), arXiv v2 numbering). This identity is also the cosmological initial condition on which the superselection theorem of §8 rests: $g_{0a} = 0$ at every epoch means, by the inversion corollary, that the universe hands every forming source a zero-charge medium. (That fluids and superfluids have massless cosmological tensors is BCP's and CCP's general statement [4, 5]; the reading for this row, the vector-freezing identity, and the consequence — the gravitomagnetic structure of §8, a vector-sector effect, rides on a tensor sector that stays GR's at every cosmological epoch — are ours.)

The protected Minkowski vacuum is the dynamical attractor of the rest of the phase's histories. The clock's equation of motion is the conservation of $J = a^3 s$, with $s = U_Y - 2\dot{\phi}U_X$ the quantity CCP identify as the medium's entropy density [5] — there a piece of thermodynamics, here the whole dynamics: expansion dilutes the density $J/a^3 \propto a^{-3}$, driving $\dot{\phi}$ to a root of $U_Y = 2\dot{\phi}U_X$; in the Minkowski basin that root is exactly the flat-space tadpole condition of §4 (the second condition, $U^* = 0$, is the usual cosmological-constant adjustment, not dynamics; the basin caveats — three roots, and a $\rho < 0$ strip of the basin needing other matter early on — are in Declared limits). Protection is not an initial-condition choice; the expanding universe relaxes the medium onto its protected corner. The mechanism — a conserved charge diluting onto a symmetry-restoration point — is the ghost condensate's [8] and DTT's [10], and we cite it as such; the identification of the attractor with *the same theory's Minkowski tadpoles* is, to our searches, new — indeed in the mirror phase it is explicitly false (DTT's attractor is not their Minkowski point [10]), and for the related vacuum-energy-degravitating superfluid of Khoury–Sakstein–Solomon the analogous question was posed in 2018 and left as future work [25]; none of its citing papers answered it (21 citing papers swept 2026-07-21; record in `campana-m2/frw/BIBLIOGRAFIA-FRW-2026-07-21.md`).

On the attractor the stress is vacuum-form; the transient is exact dust. $\rho + p = \dot{\phi}J/a^3$ holds off-shell (Bianchi verified as an identity), so the attractor has vacuum-form stress $p^* = -\rho^* = U^*$: $w = -1$ exactly whenever $U^* \neq 0$, while with the usual cosmological-constant adjustment (§4) $U^* = 0$ the endpoint is Minkowski itself and the approach is pure dust ($w \rightarrow 0$ as $\rho \rightarrow 0$ — the campaign's numerical route shows exactly this). The off-attractor component obeys $\delta\rho \propto a^{-3}$ with $\delta p = 0$ *exactly* at linear order — an algebraic identity of the class, needing only the simple-root condition $K = f'(1) = 2m_0^2 \neq 0$: the same m_0^2 of the health window $m_0^2 m_1^2 > 0$, so the attractor's good conditioning and the scalar's health are one requirement — reproducing the ghost condensate's cosmological-constant-plus-cold-dark-matter phenomenology [8, 9] — cited, not claimed. The scalar sector stays frozen on FRW: the Minkowski double zero is reproduced exactly in the flat limit, and one zero of the determinant persists exactly on the full background — provably: the ω^0 coefficient of the scalar determinant is proportional to $3H^2 - \rho$ and vanishes by the Friedmann constraint itself (the persistent zero *is* the constraint; the residual symmetry survives FRW). The rest of the attractor spectrum is **purely damped**. The v0.4–v0.5 drafts and the campaign record quoted $\omega = \pm(H/\sqrt{2})k + 3iH$ here;

this paper's own 2026-07-27 audit refuted that formula — it came from truncating the determinant one order too early. The correct leading order is

$$\det_{\text{dom}} = k^3 \cdot [-8i\omega^4 - 48H\omega^3 + 4iH^2(k^2+18)\omega^2 + 24k^2H^3\omega],$$

whose nonzero roots in the EFT window ($k \lesssim 1$ in units of Λ) are all purely imaginary in the $e^{+i\omega t}$ convention ($\text{Im } \omega > 0 = \text{decay}$): a slow diffusive mode $\omega \approx i(k^2/3)H$ and a pair $\omega = i(3 \mp k/\sqrt{2})H + O(k^2)$ — the $\pm k/\sqrt{2}$ split is real but lives in the *damping*, not the phase velocity, which is zero; a propagating pair appears only above $k^* \approx 1.27$, outside the window. Triple-checked (exact symbolic extraction; the real cubic in $y = -i\omega/H$; 60-digit numerics). The correction *strengthens* the freeze-out: no propagating mode and no instability at all in the window — every root has $\text{Im } \omega \geq 0$. (All nonzero roots have $|\omega| \lesssim 3.6H$, inside the frozen-coefficient band declared below: the numbers are indicative, the structural statement — no propagation, no instability — is the robust part.) This also answers, for this class, the published objection that $w = -1$ superfluids are exponentially unstable and need a solid component [6, 26]: our leading-order scalar is frozen (a double zero) rather than gradient-unstable, and the k^4 NLO then governs exactly as in the ghost condensate (§6) — the exponential instability of [6, 26] does not arise here (their equation lives in a different class; the .tex-level re-verification of [26] stays on the block list) — while our $w = -1$ is a dynamical fixed point, not a symmetry-imposed constraint as in [6].

Declared limits. The explicit corner's (§4) fixed-point surface has, besides Minkowski, two model-dependent extra roots — an AdS-like point unreachable as a flat-FRW endpoint (its basin recollapses) and a dS-like point with $U_X < 0$, i.e. a sick vector sector, outside the healthy corner; only the Minkowski root is structural (a linear U has it alone). The Minkowski basin of the explicit corner requires $\dot{\phi} \gtrsim 0.816$, and its $0.816 < \dot{\phi} \lesssim 0.97$ strip has $\rho < 0$ — inadmissible as the sole source in flat FRW, so histories starting there need other matter to dominate early on (declared, not cured); the ω^1 coefficient also has a second double-zero point at $w \approx 0.544$, inside the AdS basin — irrelevant to the attractor, listed for completeness. The scalar analysis uses frozen coefficients (valid for $\omega \gg H$; the $\omega \sim H$ band, where all the attractor roots live, is indicative). The full quadratic FRW action is now derived (2026-07-27): tensor and vector sectors closed as identities and the tensor-mass factor question closed as a convention (see above); the scalar sector's four-field action is built, its constraints verified, and its symbolic validation run against the corrected spectrum is in progress — that last run is the remaining FRW item on the block list. Two further internal misprints in BCP's §8 — a flipped k^2 sign in their Fourier-space GW action ((8.17), arXiv v2 numbering) and a factor-2 slip in their printed propagation equation (8.18) — were flagged in the same pass and **confirmed the same day by a dedicated two-agent verification campaign** (an independent full re-derivation from BCP's fundamental action with three routes including an exact static-solid discriminant, plus the external evidence that the authors' own follow-up JCAP 09 (2017) 036 prints both corrected forms; as printed, the $M_2^2 \rightarrow 0$ limit of (8.17) would not propagate waves). They do not affect our row ($M_2^2 \equiv 0$ identically); record and the drafted query paragraph in the campaign acta §8 (campana-m2/CIERRE-BLOCKLIST-2026-07-27.md, script in campana-m2/erratas-bcp-s8/). Both were reported to the authors on 2 Sep 2026. **L. Pilo confirmed the k^2 sign of (8.17)** (email, 15 Sep 2026 [34]); on (8.18) he remarked that the sign of the mass term depends on the overall sign of U and on the definition of m_2^2 in (7.3), and that the genuine graviton mass must enter as $\omega^2 = k^2 + m_2^2$ — with which we agree: in the variable of (7.3) that is exactly the dispersion, and the relation fixed by BCP's own (7.22) is $m_2^2 = -2M_2^2$, with M_2^2 the tensor-mass coefficient of (7.21) (the same relation as Appendix A of [5], $M_2^{\text{there}} = -2a^2 M_2^{\text{here}}$). Our point about (8.18) concerns only the coefficient of that $M_2^2 - 2M_2^2$, not M_2^2 — independently of any sign convention; it rests on the three-route verification above, and a clarification was sent to the authors (record in campana-m2/RESPUESTA-PILO-2-2026-09-15.md).

10. What this paper does not claim

- **It does not derive the Einstein–Hilbert term.** The EH kinetic term is assumed, as in the entire framework [1, 4, 5]. A fully emergent graviton faces Weinberg–Witten and needs different ideas (induced gravity, holography, or non-relativistic routes [15]); this paper does not touch that front.
- **It does not explain the hierarchy $\Lambda \ll M_{\text{Pl}}$.** Inherited from the ghost condensate, declared, not cured.
- **The scalar sector is not a discovery channel.** It is exactly the ghost condensate's (m_1^2 cancels there); every scalar-sector bound and signature of [8, 9] applies verbatim, nothing new to measure on that side.
- **The superselection theorem is conditional on its declared hypotheses (§8):** universal coupling; smooth Φ^a (no vortices — and the imported barrier physics prices vortex nucleation at $E_b \propto 1/v$, divergent at this drive); the NLO flow basis $\{\theta, \sigma, \omega, a\}$ as a structural argument, not a formal completeness theorem; $U_Y \neq 0$ with a regular metric; negligible primordial vector perturbations. If one fails — vortex nucleation is the physically sharpest candidate — the two-branch structure of §8 reopens with all its formulas intact, screened profile included. What is *not* conditional: the branch equations themselves, the charge bookkeeping, and the identity $P_a^i = 0$ of the leading-order Lagrangian (exact to all orders in $\hbar$; the NLO currents are the conditional part).
- **It does not validate the TCI program.** It answers one open question of the massive-gravity literature with one medium and its phenomenology. Within our own program the result *constrains*: by the exclusion lemma, any medium of this class underlying gravity must be positionally fluid at leading order — solids are out. (This converges with independent falsifications published in the program's record.)
- **The BCP misprint is no longer provisional:** L. Pilo confirmed it (email, 22 Jul 2026, cc D. Comelli, after re-checking their original notebooks; cited with permission [34]). Our own results never hinged on it ($U(X,Y)$ involves neither τ_n nor y_n ; the lemma's weights are the five-times-verified ones, now author-confirmed). Nothing here should be read as an endorsement of this paper's content by the BCP authors: the exchange cited in [34] concerns misprints in their own paper.

11. Open problems

(Two problems this list carried through v0.6 were closed by the state-selection campaign: state selection itself — answered by the superselection theorem of §8 — and NLO frame dragging on the co-rotating branch — answered there too: the observable exists, $\delta_{\text{ret}} = -\hat{c}(\Lambda/\bar{M}_{\text{Pl}})^2$, and is unobservable by ≥ 30 orders everywhere in the declared windows.)

1. **FRW cosmology** of $U(X,Y)$: done at background + quadratic-action level (§9); remaining: the no-ghost analysis beyond the frozen-coefficient window (the $\omega \sim H$ band, where the attractor roots live), the optional exact- δ re-run of the scalar validation, and the .tex-level re-verification of [26]'s quoted instability equation.
2. **Precise map to the tensor-order program:** the phase found here ($m_2 = m_3 = m_4 = 0, m_0^2 m_1^2 > 0$) is not identical to the phase targeted in an earlier note of our program ($m_0 = 0 \wedge m_2 = 0$); the exact relation remains to be written.
3. **Independent CAS re-derivation** (xAct/cadabra) of the dictionary — desirable redundancy (L. Pilo has since confirmed the misprints, §2 and [34]), not load-bearing.
4. **The radiative sector:** binary-pulsar damping (Hulse–Taylor) is untouched here. The tensor sector is GR's at this order, which suggests standard quadrupole radiation, but the vector/scalar bookkeeping with time-dependent sources remains to be done before any claim.
5. **Formal status of the superselection hypotheses:** a completeness theorem for the NLO flow basis $\{\theta, \sigma, \omega, a\}$ (currently a structural argument, §8), and a first-principles treatment of the vortex sector of Φ^a in

this medium (topological defects are the one escape from the smoothness hypothesis; the imported barrier argument prices them as cost-divergent at low drive, but nucleation, tension and dynamics deserve their own computation here).

Acknowledgments

The author thanks L. Pilo for re-checking the original notebooks of [4], for confirming the misprints, and for permission to cite the exchange [34]. The derivations, the code and the drafting of this paper were developed in collaboration with an AI system (Claude, Anthropic — working name *Fable*), under the author's direction. All numerical and symbolic results are reproducible with the public scripts referenced below. The author takes full responsibility for the content.

Reproducibility

All scripts are Python 3 + SymPy, in `especulativo/campana-m2/` of the public repository (github.com/IrigoyenNicolas1991/ift-theory). Individual scripts run in seconds; the heaviest (`colA_causal.py`, and `seleccion-estado/colA/a2_vertices_cubicos.py` at ~3 minutes) take a few minutes; the full suite a few minutes on a laptop. (2026-07-21: the verifier-pipeline scripts of `acople/` were repaired to run from the repository as published — import paths and two runtime bugs, listed in the audit record; the repaired suite reproduces every check quoted in this paper. The C5d check of `colA_causal.py` prints a nonzero residue that was reconciled on 2026-07-27: an artifact of that check's own extraction method, not physics — `acople/estatico/C5d_dirimido.py`, 26 exact checks; see block-list item 7.)

Block	Scripts	What they verify
Dictionary + calibrations	<code>masas_medio.py</code> , <code>calibracion_fonones.py</code>	general dictionary; isotropy; BCP (7.5); Dubovsky $m_1=0$ phase; gold calibration (Goldstone route, remainder 0); transverse sector
U(X,Y) and the protected phase	<code>parteB_medioXY.py</code>	off/on-shell masses; ghost-free corner; Schur identity; nonlinear invariance; residual-symmetry invariance; exclusion lemma
Misprint verification	<code>verificacion_independiente.py</code> , <code>refutador_pesos_bcp.py</code>	exact-inverse rederivation; blind refutation (4 methods)

Block	Scripts	What they verify
Scalar sector	<code>escalar/derivador/*</code> , <code>escalar/verificador/*</code> , <code>escalar/cotas_fenomenologicas.py</code>	EH from scratch; LO freeze-out; NLO dispersion; ACLM dictionary; bounds — derivation and independent verification pipelines share no code
Matter coupling	<code>acople/estatico/*</code> , <code>acople/frame-dragging/*</code> , <code>acople/verificador/*</code>	causal theorem; $\gamma(p)$; two branches; rotating-sphere exterior; precession factors; bounds — again with a disjoint verification pipeline
C5d reconciliation	<code>acople/estatico/C5d_dirimido.py</code>	the C5d residue as an extraction artifact (symbolic identity); physical dispersion by Routhian and 4×4 determinant = scalar-pipeline formula; $\partial \omega^2 / \partial m_1^2 = 0$; $\rho_{\text{op}} \equiv \rho$ term by term
Completeness of $\{X, Y\}$	<code>exhaustividad_XY.py</code>	chain-rule pointwise reduction; normal form and explicit connecting group elements; orbit count $10-8 = 2$; symbolic Schur; candidate invariants dismissed (incl. Z_ϵ); $\text{sl}(3)$ calibration (recovers BCP's superfluid row); no-clock corollary

Block	Scripts	What they verify
FRW (§9)	frw/fondo_frw.py, frw/verificador/v1_fondo_covariante.py – v4b_escalar_frw_analisis.py, frw/accion_cuadratica_frw.py, frw/factor_722.py	three-route background; FLRW tensor identity + control against BCP's mass formula (their (7.21), arXiv v2 numbering); fixed- point map and basins; frozen- coefficient scalar analysis; full quadratic action (tensor/vector closed as identities, 17- check convention audit of the tensor-mass factor; scalar validation confirmed the corrected spectrum, see block list 5)

Block	Scripts	What they verify
State selection (§8)	seleccion-estado/colA/a1_teorema_via_schur.py – a4_numeros.py, seleccion-estado/colB/t1_carga_noether.py – t4_robustez.py, seleccion-estado/verificador/va1_teorema.py – va4_numeros.py	the superselection identity $P_a^i = 0$ by four routes (solve-u; Levi-Civita; Gram determinants; vertex-by-vertex — 36 cubic components zero); inversion corollary; constraint $C = -m_1^2 S$; universal IVP incl. moving sources and non-axisymmetric rotation; no compensating-cloud solution; sector energetics; two-mode NLO no-leak; δ_{ret} with generic acceleration coefficient (no order reduction); pole-free transfer function; leak and spin-down numbers — the verifier pipeline (<code>va*</code>) shares no code or parametrization with the columns

Novelty search (method declared)

Full-text INSPIRE searches (2026-07-21): "unrestricted internal diffeomorphisms" — 0 hits; "unrestricted spatial diffeomorphisms" — 3 hits, all in shape dynamics/unimodular gravity (physical-space diffeomorphisms, different object); "shear rigidity" $\wedge$ "graviton mass" and "graviton mass" $\wedge$ "shear modulus" — the holographic-elasticity literature [17] and crystal gravity [18], which relate mass and rigidity in other frameworks but do not state the protection lemma in the BCP/Dubovsky setting; Lense–Thirring $\wedge$ massive gravity and gravitomagnetic $\wedge$ Meissner sweeps — no screened frame dragging in any massive/LV gravity phase (closest echo: [19], laboratory superconductors). All 19 papers citing [1] since 2024 were listed; none constructs a medium for the protected phase. The 2026-07-20 campaign additionally swept arXiv/INSPIRE 2004–2026 for the $U(X,Y)$ identification (closest: CCP's isentropic $U(X+Y^2)$ [5], without the symmetry or the phase connection). The full search log — queries verbatim, hit lists, the 19 citing papers, and declared limitations — is recorded in `campana-m2/BARRIDO-INSPIRE-2026-07-21.md`. The FRW novelty claims of §9 rest on the 2026-07-21 FRW bibliographic campaign: the 21 papers citing [25] and the 34 citing [5] were listed and checked (none does the $U(X,Y)$ cosmology; KSS's 2018 question stands unanswered), plus full-text sweeps for cosmological-attractor $\wedge$ massive-gravity variants; record in `campana-m2/frw/BIBLIOGRAFIA-FRW-`

2026-07-21.md. The completeness claim of §3 rests on the 2026-07-27 pass: full-text checks of [1] (eqs. (14)–(15), stops at the mass pattern), [4] ((4.9), §5.1.2, $\det = 1$ imposed), [7] ($\det = 1$), and NPPR arXiv:1501.03845 (§2.1, stops at $\text{Diff}^{\prime}(3)$), plus web sweeps for the unrestricted case — no prior classification found. The state-selection claims of §8 rest on the 2026-07-27 bibliographic column (record in `campana-m2/seleccion-estado/BIBLIOGRAFIA-SELECCION-2026-07-27.md`, primary sources read at the .tex level, exceptions declared in the record): a dedicated INSPIRE sweep (seven queries declared there) found no discussion of metastability or state selection of Stückelberg-medium configurations around rotating sources in any $m_1^2 \neq 0$ phase; the moving-source problem is solved in the literature only for the irrotational scalar sector [29, 30, 9]; the persistent-current barrier framework is imported from [32]; the nearest-headline result [31] uses a different mechanism (static spatial gradient, no rotation, no vector sector). If prior art exists that these searches missed, the novelty rows above are forfeit and this file will say so.

References

(Bibliographic data verified against INSPIRE-HEP / primary sources, 2026-07-27; the record of that verification pass, including the [16]–[18] content checks, lives in `campana-m2/CIERRE-BLOCKLIST-2026-07-27.md`.)

- [1] S. L. Dubovsky, "Phases of massive gravity", JHEP 10 (2004) 076, arXiv:hep-th/0409124.
- [2] V. A. Rubakov, "Lorentz-violating graviton masses...", arXiv:hep-th/0407104.
- [3] V. A. Rubakov, P. G. Tinyakov, "Infrared-modified gravities and massive gravitons", Phys. Usp. 51 (2008) 759, arXiv:0802.4379.
- [4] G. Ballesteros, D. Comelli, L. Pilo, "Massive and modified gravity as self-gravitating media", Phys. Rev. D 94 (2016) 124023, arXiv:1603.02956.
- [5] M. Celia, D. Comelli, L. Pilo, JCAP 09 (2017) 036, arXiv:1704.00322.
- [6] M. Celia, D. Comelli, L. Pilo, "Self-gravitating Λ -media", JCAP 01 (2019) 057, arXiv:1712.04827.
- [7] M. Celia, D. Comelli, L. Pilo, R. Rollo, "Adiabatic Media Inflation", JCAP 12 (2019) 018, arXiv:1907.11784 (appendix A mass formulas used in §2).
- [8] N. Arkani-Hamed, H.-C. Cheng, M. A. Luty, S. Mukohyama, "Ghost condensation...", arXiv:hep-th/0312099.
- [9] N. Arkani-Hamed, H.-C. Cheng, M. A. Luty, S. Mukohyama, T. Wiseman, "Dynamics of gravity in a Higgs phase", JHEP 01 (2007) 036, arXiv:hep-ph/0507120 (CMB "twinkling" bound; ghost dark matter).
- [10] S. L. Dubovsky, P. G. Tinyakov, I. I. Tkachev, "Cosmological attractors in massive gravity", Phys. Rev. D 72 (2005) 084011, arXiv:hep-th/0504067.
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- [14] D. T. Son, "Effective Lagrangian and topological interactions in supersolids", Phys. Rev. Lett. 94 (2005) 175301, arXiv:cond-mat/0501658.
- [15] L. Chojnacki, R. Pohle, H. Yan, Y. Akagi, N. Shannon, "Gravitational wave analogs in spin nematics and cold atoms", Phys. Rev. B 109 (2024) L220407, arXiv:2310.10078.

- [16] S. Endlich, A. Nicolis, J. Wang, "Solid Inflation", JCAP 10 (2013) 011, arXiv:1210.0569.
- [17] L. Alberte, M. Baggioli, A. Khmelnitsky, O. Pujolàs, "Solid Holography and Massive Gravity", JHEP 02 (2016) 114, arXiv:1510.09089 (and the holographic-elasticity literature that followed).
- [18] J. Zaanen, F. Balm, A. J. Beekman, "Crystal gravity", SciPost Phys. 13 (2022) 039, arXiv:2109.11325.
- [19] C. J. de Matos, M. Tajmar, "Gravitomagnetic London moment and the graviton mass inside a superconductor", Physica C 432 (2005) 167, arXiv:cond-mat/0602591.
- [20] LIGO-Virgo-KAGRA, GWTC-3 tests of GR (graviton-mass bound $m_g \leq 1.27 \times 10^{-23}$ eV), arXiv:2112.06861.
- [21] I. Ciufolini et al., "A test of general relativity using the LARES and LAGEOS satellites and a GRACE Earth gravity model", Eur. Phys. J. C 76 (2016) 120, arXiv:1603.09674.
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- [22] L. Iorio, "Limitations in Testing the Lense–Thirring Effect with LAGEOS and the Newly Launched Geodetic Satellite LARES 2", Universe 9 (2023) 211, arXiv:2304.14649; and "Will LAGEOS and LARES 2 succeed in accurately measuring frame-dragging?", Eur. Phys. J. C 85 (2025) 255, arXiv:2503.07264.
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- [30] M. Peloso, L. Sorbo, "Moving sources in a ghost condensate", Phys. Lett. B 593 (2004) 25, arXiv:hep-th/0404005.
- [31] V. V. Kiselev, "Ghost condensate model of flat rotation curves", arXiv:gr-qc/0406086 (unpublished, to our records).
- [32] E. Varoquaux, "Anderson's considerations on the flow of superfluid helium: Some offshoots", Rev. Mod. Phys. 87 (2015) 803, arXiv:1406.5629 (persistent-current stability, phase-slip barriers, and the near-threshold caveats used in §8).
- [33] D. Krotov, C. Rebbi, V. Rubakov, V. Zakharov, "Holes in the ghost condensate", Phys. Rev. D 71 (2005) 045014, arXiv:hep-ph/0407081.
- [34] L. Pilo, private communication (emails to the author, 22 Jul 2026 and 15 Sep 2026, cc D. Comelli): confirmation, after re-checking the original notebooks, of the misprints in eqs. (7.6)–(7.7) of [4], and of the k^2 sign in eq. (8.17) of [4]; permission to cite the exchange granted on 15 Sep 2026.

Pre-publication block list (nothing leaves the repo before these)

- 1. In-house adversarial audit of THIS TEXT — three passes DONE** (2026-07-21: 31 findings applied in v0.2, then 12 more in v0.3, record in `campana-m2/AUDITORIA-TEXT0-PAPER-2026-07-21.md`; 2026-07-27: third pass on the full v0.5 text — 30 findings, applied in v0.6 together with the §9 physics audit — record in `campana-m2/CIERRE-BLOCKLIST-2026-07-27.md`). The **fourth text-audit pass over the v0.7 rewrite ran the same day** (2026-07-28, two independent agents: an adversarial text audit — 31 findings: 4 high, 11 medium, 16 low — and a fidelity check of every number, formula, attribution and reference of the new text against the campaign records — 7 findings; all applied except those explicitly deferred, record in `campana-m2/AUDITORIA-TEXT0-v0.7-2026-07-28.md`; the state-selection integration touched the title, abstract, scoreboard, §§1, 3, 6, 7–11, Reproducibility, Novelty search and references). Remaining: one final short read of the assembled text before submission, and a ~250-word submission abstract (the current one is an extended summary, several times arXiv's metadata limit — flagged by the fourth audit, deferred to the channel decision).
- 2. Experimental accuracies — closed to the paywall line 2026-07-27:** primary references cited ([21]–[24], [28]) with complete bibliographic data (INSPIRE pass; two corrections caught: [19] author order, [24]'s second item is EPJ Plus). [23] verified against its openly served primary material — Extended Data Tables 1–3 (the 0.2% Total-RSS, $\mu = 1.0001 \pm 0.0019$, semi-major axes), Supplementary, and the 88-page Peer Review File (which contains the quantitative reply to the injection-accuracy objection). Remaining: the paywalled main text of [23] (pp. 332–335, no preprint exists); full titles for [2], [5]; journal data for [8], [25], [26] if published.
- 3. BCP misprint — RESOLVED 2026-07-22; citation permission GRANTED 2026-09-15:** the letter was sent by the author on 21 Jul 2026 and L. Pilo confirmed the misprints the next day (email, cc D. Comelli), agreeing with our expressions; permission to cite the exchange was granted on 15 Sep 2026 (reference [34]), together with the confirmation of the k^2 sign of (8.17). Remaining on our side: nothing; a PRD erratum is their call; an xAct/cadabra run remains desirable redundancy.
- 4. Reference data — RESOLVED 2026-07-27** (INSPIRE-HEP verification pass, record in `campana-m2/CIERRE-BLOCKLIST-2026-07-27.md`): [7] verified in full — authors, journal, and content (its appendix-A mass formulas carry exactly the weights our §2 attributes to it, flat in U_{y_n} including $n = 0$, n^2 in U_{τ_n} , signs included); journal data completed for [6], [10]–[15], [17]–[19], [21]–[24]. Residual items folded into item 2.
- 5. FRW section — WRITTEN 2026-07-21, AUDITED AND UPGRADED 2026-07-27.** The 2026-07-27 adversarial physics audit refuted one leading-order formula (the attractor dispersion — corrected in §9 with the refutation declared; correction addendum in `frw/FRW-2026-07-21.md`; the correction strengthens the freeze-out) and confirmed the rest by independent re-derivation. The full quadratic FRW action was then derived (`frw/accion_cuadratica_frw.py`): tensor and vector closed as FLRW identities (vector: new result, three routes + BCP's vector Lagrangian, their (7.23), term-by-term on-shell), and the tensor-mass global factor closed as a convention (`frw/factor_722.py`, 17 checks; BCP's printed mass formula — their (7.21), arXiv v2 numbering — is correct). Audit-trail note, declared because the trail matters: the quadratic-action agent re-numbered our (7.21) citations to "(7.22)" by counting .tex environments; the same day's external-evidence pass and direct inspection of the arXiv PDF showed that renumbering was itself wrong — (7.21) is the mass formula, (7.22) the tensor action — and all citations now carry the arXiv v2 numbering explicitly (the journal PDF remains inaccessible to us). **The scalar validation run finished the same day and CONFIRMS the corrected spectrum structurally:** exact Minkowski limit (double zero, $\text{coef}(\omega^4) \propto m_0^2 m_1^2$); persistent zero = Friedmann constraint confirmed from the action; all roots purely imaginary in the EFT window (zero phase velocity); exact root match with the audit's cubic at $k = 1$; and the new determinant is $\propto$ the v4 determinant with an ω -independent ratio, so the audit's

exact- δ 60-digit run transfers to it (transitivity). Three surviving detail-level FAILs have their cause identified in the script's own code — it substitutes the linearized attractor $w = 1 + (3/2)H^2$, whose $O(H^4)$ error feeds the determinant's constant term at leading order, the same class of trap that produced the original dispersion error; the spurious term $\propto k^2(k^2-1)$ vanishes at $k = 1$, where everything matches exactly. Full breakdown, script flags, and the optional formal redundancy (re-run block S5 with exact δ) in CIERRE-BLOCKLIST-2026-07-27.md §6. **Remaining:** that optional exact- δ re-run, and the .tex-level re-verification of [26]'s quoted instability equation. The radiative sector stays open and named (§7, §11.4).

6. **Explicit OK from Nicolás** on: channel (arXiv hep-th + PRD/JCAP as per the campaign plan), timing, and the final text. House policy: nothing public without per-piece sign-off.
7. **C5d check — RESOLVED 2026-07-27** (acople/estatico/C5d_dirimido.py, 26 symbolic checks, ~ 10 s). The nonzero residue was an artifact of the check's own extraction method, not physics: with $\rho_{\text{op}} \neq 0$ the $\text{tr } \tilde{K} \cdot R3$ operator couples ψ to $\dot{E}$ in the reduced Lagrangian, so reading ω^2 off the raw $\ddot{\psi}$ coefficient omits exactly $-G^2/(4AF-D^2)$ — reproduced symbolically as the printed residue. The physical dispersion of the same Lagrangian (Routhian at conserved Π_E , and the 4×4 determinant — colA_cierre.py D1) equals the scalar-pipeline formula exactly, with $\partial\omega^2/\partial m_1^2 = 0$: the exact m_1^2 cancellation is confirmed from the coupling pipeline too. Bonus closure: $\rho_{\text{op}} \equiv \rho$ (same operator, same normalization, term-by-term check [7a]/[7b]), so §6's former non-identification caveat is dropped.
8. **Characterizations of [16]–[18] — RESOLVED 2026-07-27**: all three verified against their source texts (record with textual quotes and locations in campana-m2/CIERRE-BLOCKLIST-2026-07-27.md). All three RESPALDA: [16]'s tensor mass term $m_Y^2 = 4\epsilon H^2 c_T^2$ with c_T^2 carrying the solid's moduli (vanishing for the perfect fluid), celebrated there as the blue-tilt signature; [17] encodes rigidity "exclusively in the $m_2(r)$ mass parameter" (same name, same role as our lemma); [18] derives $m_G = \sqrt{(16\pi G\mu/c^2)}$ with μ the shear modulus, gravitons coupling "exclusively to the shear stress".
9. **Completeness of {X, Y} — RESOLVED 2026-07-27**: normal-form classification proven and machine-verified (campana-m2/exhaustividad_XY.py, 19/19 checks in exact arithmetic; full statement now in §3): $GL(3)$ orbits = fibers of $(X, W = X + Y^2)$ on the non-degenerate domain; the ϵ -Jacobian Z_ϵ is not invariant; $sl(3)$ calibration recovers BCP's superfluid row $\{X, Y, b\}$; no-clock corollary (a medium of Φ^a alone with the unrestricted symmetry is dynamically empty). No prior classification of the unrestricted case found ([1] stops at the mass pattern; [4], [7] impose $\det = 1$; NPPR 1501.03845 §2.1 stops at $\text{Diff}'(3)$); methodological precedent cited: [27], App. A. Declared scope: leading order only; Wess–Zumino terms outside.
10. **State selection — RESOLVED 2026-07-28 (the superselection theorem, §8)**: campaign of three independent columns plus a final adversarial verifier with its own pipeline (third parametrization of Y , first-order $\Gamma\Gamma$ Einstein–Hilbert; five attack flanks, none broke it); pre-declared decision criteria; records in campana-m2/seleccion-estado/ (SINTESIS-SELECCION-2026-07-28.md, verificador/VERIFICADOR-FINAL-2026-07-28.md; the corrected sentences of the coupling record carry a dated addendum in campana-m2/ACOPLE-MATERIA-PPN-2026-07-21.md). Known, declared discrepancy kept as-is for the record: the closing print statement of seleccion-estado/colA/a4_numeros.py says "6e-14 / 7e-36" where its own table (and the verifier's recomputation) give $2.7 \times 10^{-14} / 3.5 \times 10^{-35}$ — the table is what this paper quotes; the script is left untouched as record. Declared weakness, stated in §8: the leak estimate is a dimensional chain with ≥ 14 orders of cushion, not a vertex computation. Remaining: nothing load-bearing; the hypotheses' formal status is open problem §11.5.